Homework 1

Due: Oct. 8, 11:59 am

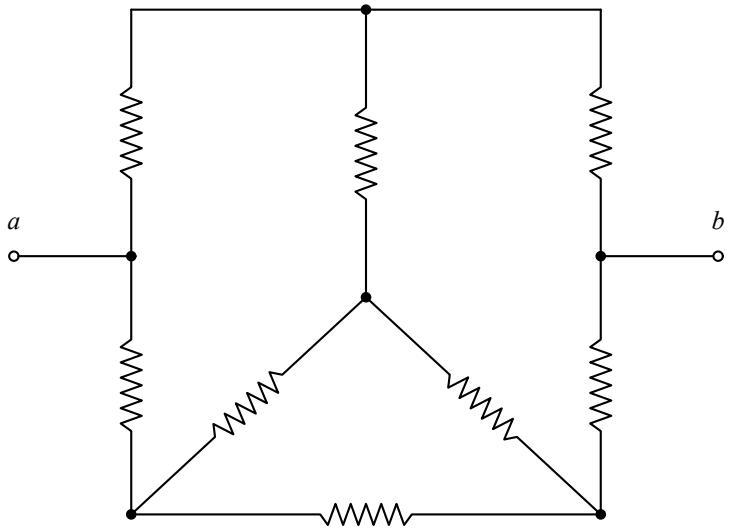
Submit your homework outside the instructor's office:
Room 324, Building 3, SIST

Rules

- Please work on your own. Discussion is permissible, but extremely similar submissions will be judged as plagiarism.
- Please show all intermediate steps: a correct solution without an explanation will get zero credit.
- Please submit on time. No late submission will be accepted.
- Please prepare your submission in English only. No Chinese submission will be accepted.
- Please retain 2 decimal places if rounding is required (unless otherwise specified).

1

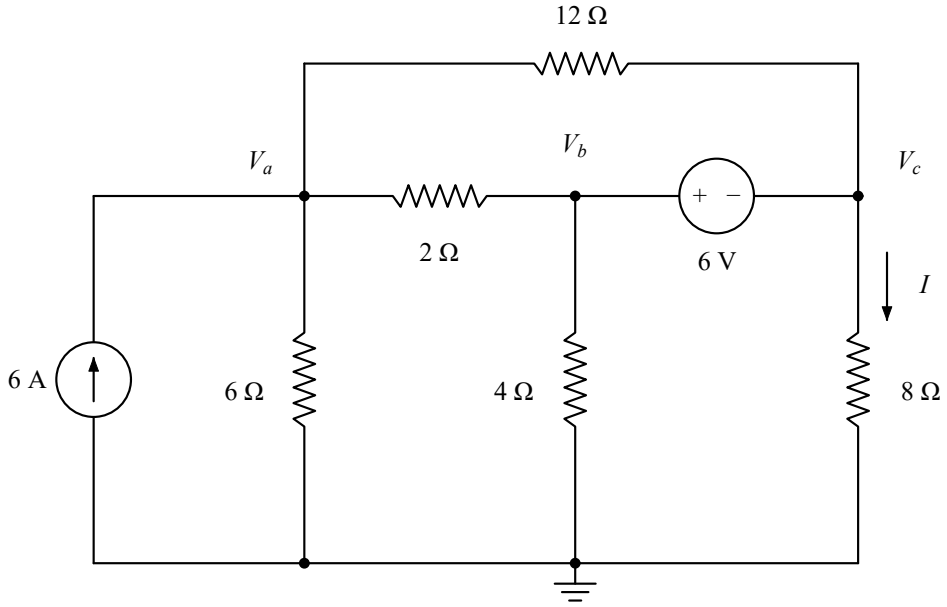
Each resistor has a resistance of $10\,\Omega$. Use a Delta $\leftrightarrow$ Y transformation to calculate the equivalent resistance R_{ab}.



All resistors: 10 Ω

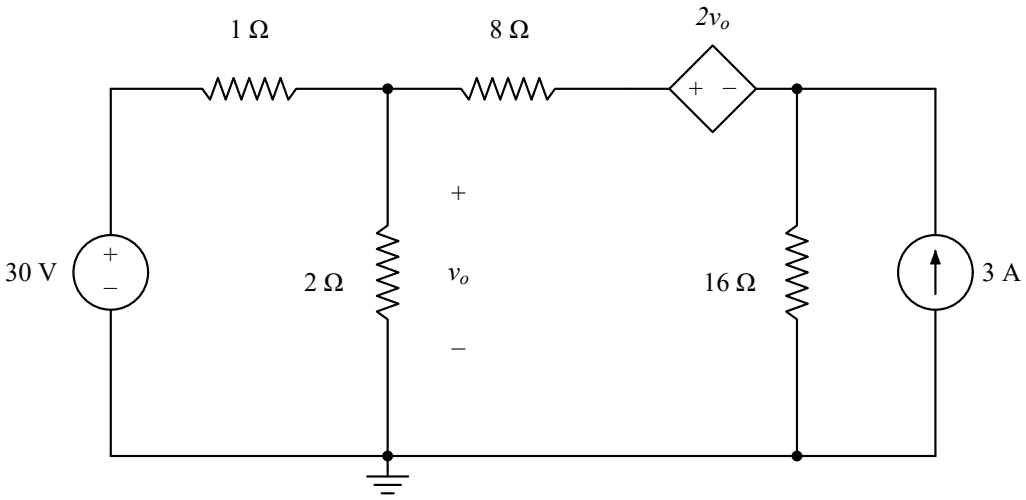
2

Use nodal analysis to find V_a, V_b, V_c, and I. Find the power absorbed by each element and verify that the total power is zero.



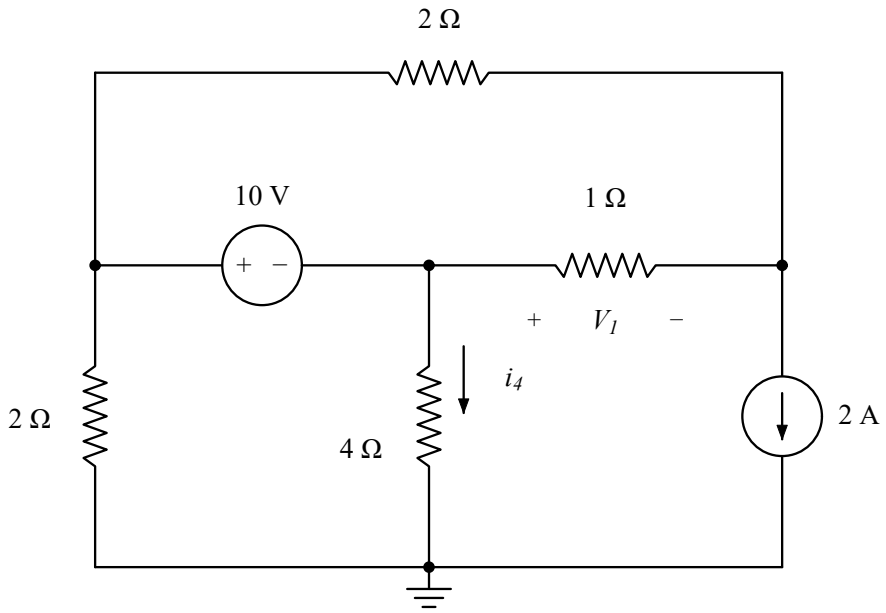
3

Use nodal analysis to find v_o in the circuit below.



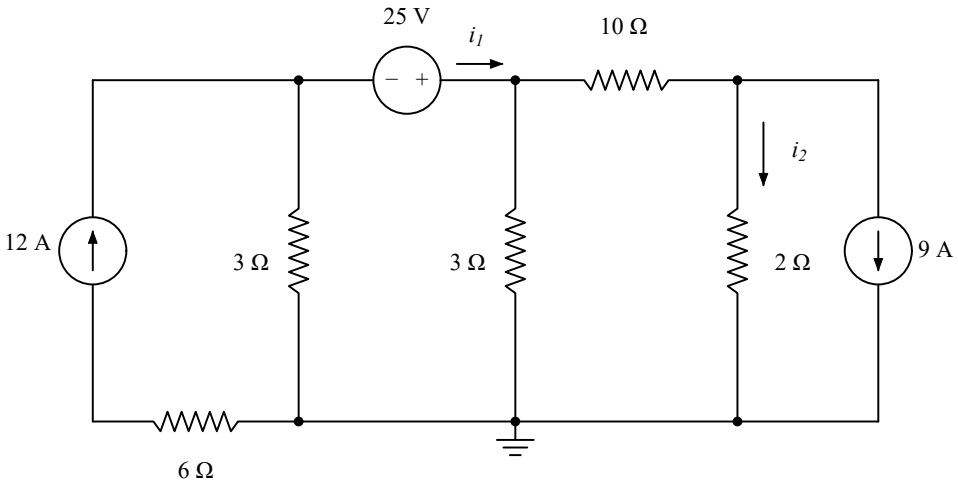
4

Calculate the branch current through the 4 Ω resistor and the voltage V_1 in the circuit below using (a) the node-voltage method and (b) the mesh-current method.



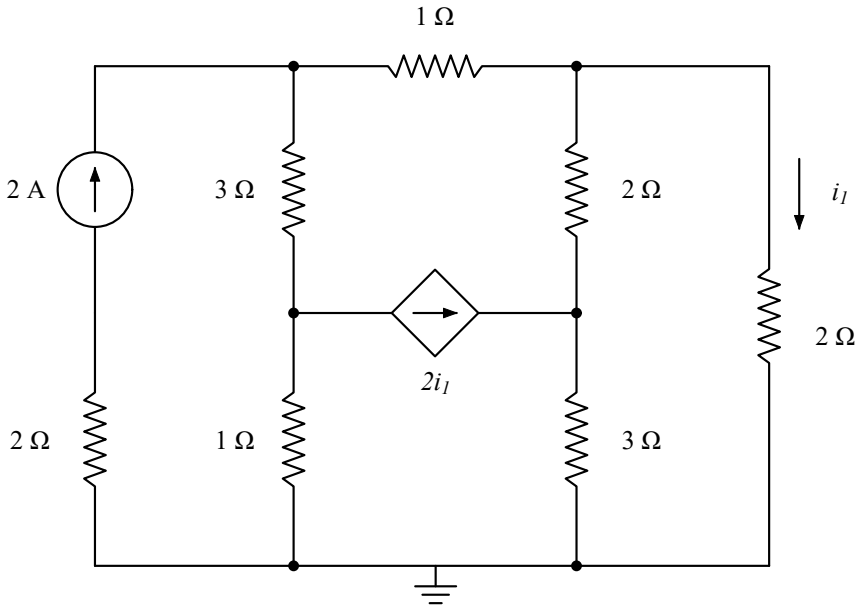
5

Use either nodal analysis or mesh analysis to find i_1 and i_2.



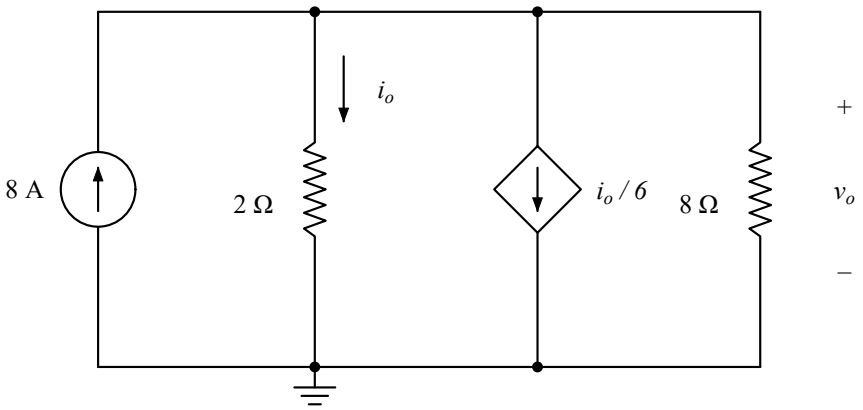
6

Use the mesh current analysis method to find i_1.



7

Use the mesh current analysis method to find v_o and i_o in the circuit below.



8

Use the mesh current analysis method to find the node voltages V_1, V_2, and V_3 for the circuit below, relative to the indicated ground.

